ROB545 Final Project Proposal

Bryan Olmstead, Nabin Gajurel

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Introduction and Motivation for Final Project

Visual Robotics, a startup company Bryan founded in Eugene, makes vision-guided robotic workcells. Their newest product uses the UFactory-850 robot arm and a Luxonis OAK4D camera mounted on the wrist of the robot. The camera detects moving objects in 3D space in 4 degrees of freedom (x, y, z, θ) to guide the robot for pick-and-place operations from conveyors. Objects are detected via template-matching from the camera's 2D image and are localized in 3D space using average depth within the template region measured with its stereo camera, both with known extrinsic calibration to the wrist of the robot. Software for this system uses ROS2 and Gazebo for simulation. Prior products used a bespoke visual servoing algorithm. It is desirable to use a more optimized algorithm for this new product.

Project Description

Implement Position-Based Visual Servoing (PBVS) as described by Corke (1996) in *Visual Control of Robots: High-Performance Visual Servoing*. Specifically, we plan to implement the estimated feedforward compensation method illustrated in Figure 1, reproducing Figure 7-20 from Corke (1996).

A programmatically defined moving object and a UFactory-850 robot will be simulated in Gazebo and controlled in ROS2. The visual servoing algorithm will be implemented in Python, receiving position from the robot and actuating it in order to intercept or track a moving object. This project uses the defined model of the object's location as a proxy for the recovered position determined from processing a captured image.

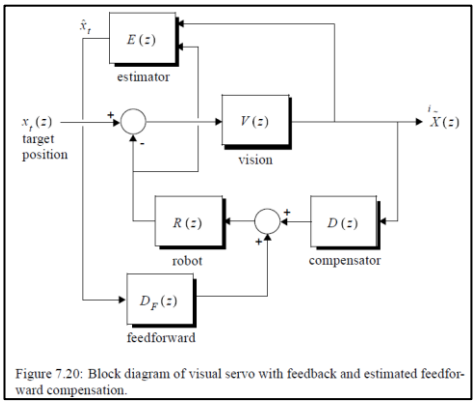


Figure 7.20: Block diagram of visual servo with feedback and estimated feedforward compensation.

Figure 1 - Algorithm

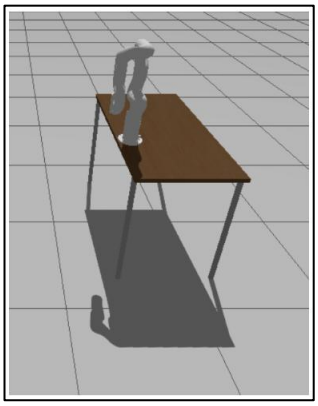


Figure 2 – Gazebo Sim

Project Phases

1. Working Gazebo simulation environment with robot control via ROS2.
2. Generation of moving object in Gazebo.
3. Implement visual servoing algorithm in Python.
4. Project presentation animation, slides, and report.